

CompositionMeasure for zCDP

Michael Shoemate (@Shoeboxam)

This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for zCDP in **mod.rs** at commit **f5bb719** (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 class CompositionMeasure(zCDP):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         match adaptivity:
6             case Adaptivity.FullyAdaptive:
7                 return Composability.Sequential
8             case _:
9                 return Composability.Concurrent
10
11     def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
12         d_out = 0.0
13         for d_mid in d_mids:
14             d_out = d_out.inf_add(d_mid)
15         return d_out
```

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` `adaptivity` and `out-composability`. Otherwise returns an error.

Definition 1.2 (Definition of zCDP). For any two distributions Y, Y' and any non-negative d , Y, Y' are d -close under the zero-concentrated divergence measure if, for every possible choice of $\alpha \in (1, \infty)$,

¹See new changes with `git diff f5bb719..6ff73bc0 rust/src/combinators/sequential_composition/mod.rs`

$$D_\alpha(Y, Y') = \frac{1}{1-\alpha} \mathbb{E}_{x \sim Y'} \left[\ln \left(\frac{\Pr[Y=x]}{\Pr[Y'=x]} \right)^\alpha \right] \leq d \cdot \alpha. \quad (1)$$

Lemma 1.3. 1. $\mathcal{F}(\rho_1, \rho_2, \dots; \rho) = \mathbb{I}(\sum_i \rho_i \leq \rho)$ is a valid ρ -zCDP IM-filter.

2. $\mathcal{G}(\rho_1, \dots, \rho_k) = \sum_{i=1}^k \rho_i$ is a valid ρ -zCDP IM-privacy loss accumulator.

Proof. For any fixed choice of $\alpha > 1$, we have that $\mathcal{F}(\rho_1, \rho_2, \dots; \rho) = \mathbb{I}(\sum_i \epsilon_i / \alpha \leq \epsilon / \alpha)$, by 1.2, which by Theorem 1.22[VW21] is a valid (α, ϵ) -RDP IM-filter.

Similarly, for any fixed choice of $\alpha > 1$, we have that $\mathcal{G}(\rho_1, \dots, \rho_k) = \sum_i \epsilon_i / \alpha \leq \epsilon / \alpha$, by 1.2, which by Theorem 1.22[VW21] is a valid (α, ϵ) -RDP IM-privacy loss accumulator. $\square$

Proof. By the postcondition of **InfAdd** we have that $\sum_i \mathbf{d_mids}_i \leq \mathbf{compose}(\mathbf{d_mids})$.

Adaptivity	Sequential	Concurrent
Non-Adaptive	Lemma 1.7[BS16]	Corollary 1[Lyu22]
Adaptive	Lemma 1.7[BS16]	Corollary 1[Lyu22]
Fully-Adaptive	Remark 4.4[FZ22]	1.3

This table is reflected in the implementation of **composability** on line 2.

$\square$

References

- [BS16] Mark Bun and Thomas Steinke. Concentrated differential privacy: Simplifications, extensions, and lower bounds, 2016.
- [FZ22] Vitaly Feldman and Tijana Zrnic. Individual privacy accounting via a renyi filter, 2022.
- [Lyu22] Xin Lyu. Composition theorems for interactive differential privacy, 2022.
- [VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.